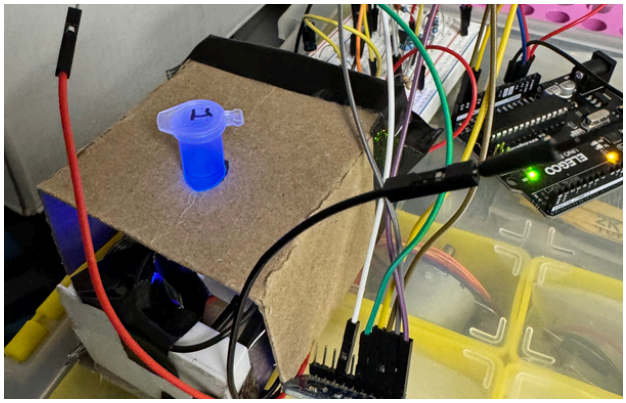
LUKA SKARO

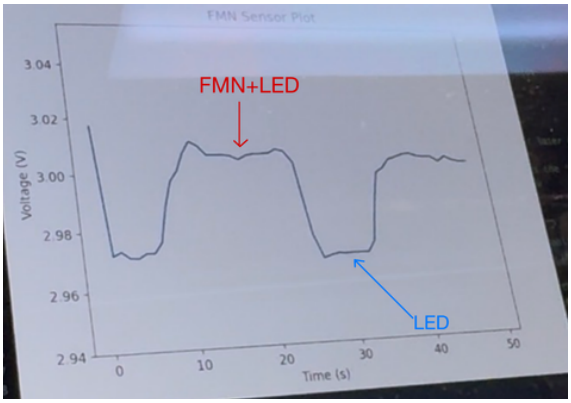
NANOTECHNOLOGY ENGINEERING STUDENT

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OPTICAL BIOSENSOR - HEPATOPHORE™



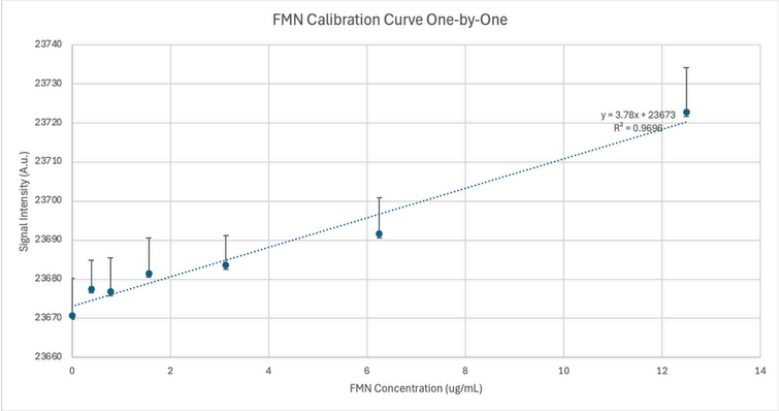


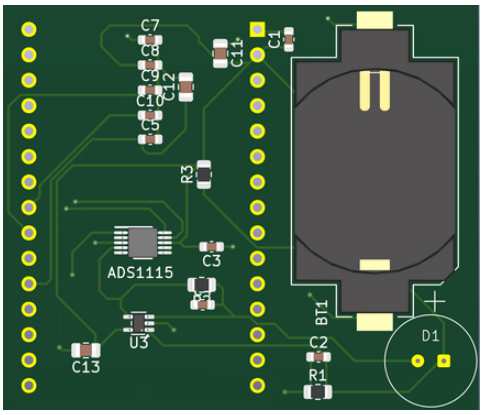
What?

- Designed and built a sensor that detects micromolar concentrations of a biomarker during liver transplantation to assess organ viability.
- Provides a direct and rapid method for assessing ischemia-reperfusion injury without bulky equipment such as spectrophotometers.

How?

- Designed and simulated a prototype circuit in LTspice, including a photodetector → TIA → microcontroller signal chain.
- Tested the live circuit on a breadboard using diluted biomarker concentrations, as well as biomarker-spiked human blood plasma.





Results

- Calibration curves across various biomarker concentrations yielded R ≈ 0.97, indicating a near-linear relationship in saline solution.
- The optical biosensor successfully detected the biomarker in human plasma.
- Sensor calibration curves yielded results similar to spectrophotometry measurements from clinical data.

Future

- Build a robust PCB using KiCad and collaborate with manufacturers in China for fabrication.
- Future versions will use an ESP32 architecture for seamless Bluetooth connectivity.
- Transition to live monitoring during normothermic regional perfusion by integrating the sensor with the perfusate circuit during transplantation.

LUKA SKARO

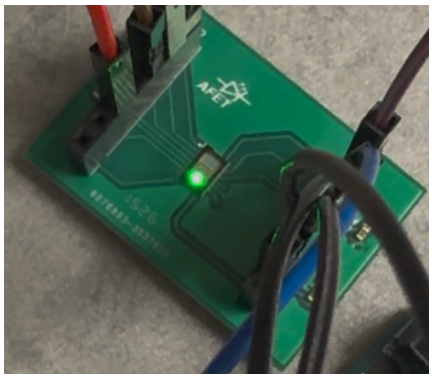
NANOTECHNOLOGY ENGINEERING STUDENT

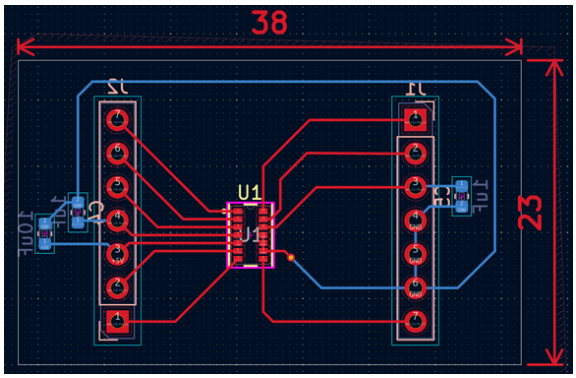
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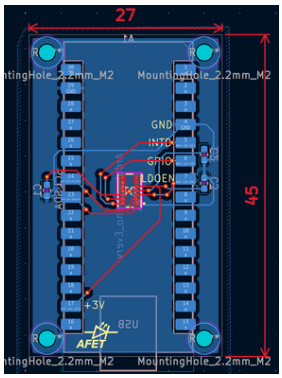
WEARABLE PPG SENSOR

AFET Advanced Flexible Electronics Technology

UNIVERSITY OF WATERLOO







What?

- Designed a wearable PPG (photoplethysmography) sensor that measures oxygen saturation (SpO_2) and heart rate.

How?

- Used KiCad to design a PCB incorporating the PPG sensor.
- Integrated the PCB → Arduino Nano → Python GUI to control LED colour, intensity, and duty cycle.
- Worked with manufacturers in China to fabricate the PCBs.

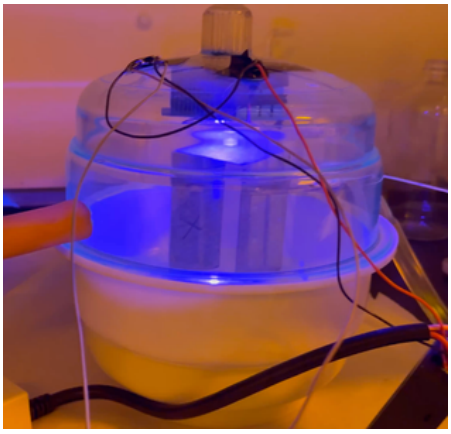
Results

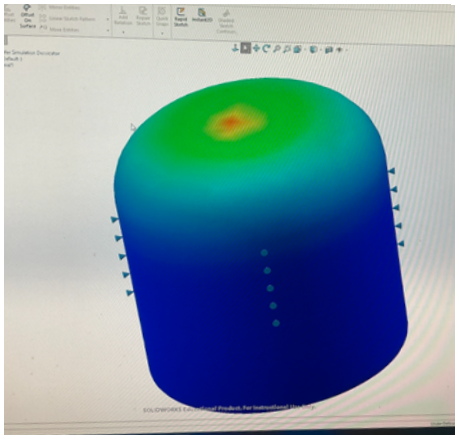
- The PPG sensor successfully measures heart rate and SpO_2.
- Established an active partnership with university and healthcare laboratories to validate the PPG sensor.
- Current design integrates Microcontroller and sensor

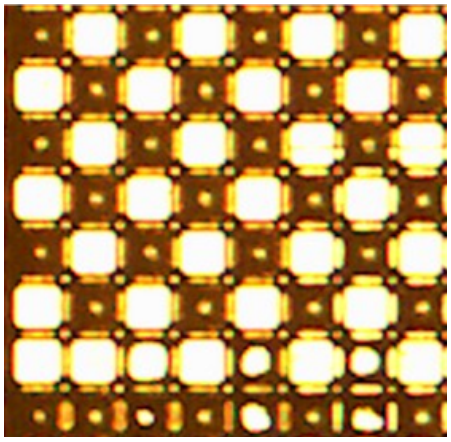
QUANTUM DOT CURING VESSEL

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What?

- Converted an existing desiccator into a reaction vessel for curing quantum-dot-containing polymers.
- A UV-A LED cures the polymer matrix, suspending quantum dots within microwells that serve as display pixels.

How?

- Used FEA thermal analysis in SolidWorks to determine the required heatsink dimensions for cooling the LED.
- Created an N_2 environment by adding an inlet fitted with a grommet to limit unwanted side reactions.
- Controlled LED intensity and duty cycle using an NMOS transistor and microcontroller.

Results

- Rapid and precise curing produced a fully cross-linked polymer–quantum dot matrix with well-defined microwells for quantum-dot displays.